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Bihar Engineering University, Patna

B.Tech. 6th Semester Examination, 2023

Time: 03 Hours

Code: 101605 ***Subject: Environmental Engineering-II*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose/Write the correct answer of the following (Any seven question only): **[14]**

(a) Imhoff cone is used to determine
(i) settleable solids
(ii) suspended solids
(iii) dissolved solids
(iv) None of these

(b) The main objective of secondary treatment of wastewater is
(i) COD removal
(ii) BOD removal
(iii) nutrient removal
(iv) All of the above

(c) The ratio of maximum daily sewage flow to the average daily sewage flow for sewer mains of 0.5 to 1 m diameter, is generally taken as
(i) 1.5
(ii) 2.0
(iii) 3.0
(iv) None of these

(d) The percentage of activated sludge mixed with the wastewater in activated sludge process of wastewater treatment ranges between
(i) 10 to 20
(ii) 20 to 40
(iii) 40 to 60
(iv) None of these

(e) The bacteria which require free oxygen for their survival are called
(i) aerobic bacteria
(ii) anaerobic bacteria
(iii) facultative bacteria
(iv) None of these

(f) The most serious environmental effect posed by hazardous waste is
(i) air pollution
(ii) contamination of ground water
(iii) increased used of land for landfills
(iv) None of these

(g) The most suitable section of a sewer in a separate sewerage system is:
(i) rectangular
(ii) circular
(iii) new egg shape
(iv) parabolic

(h) The gas which is generally found present in sewers is:
(i) H_2S
(ii) CO_2
(iii) CH_4
(iv) all of these

(i) One of the most effective forms of hazardous waste disposal may be
(i) land fills
(ii) deep well injection
(iii) Incineration
(iv) Surface impoundments

(j) Which one of the following solid waste disposal methods is ecologically most acceptable?
(i) Sanitary landfill
(ii) Incineration
(iii) Pyrolysis
(iv) Composting

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Q.2 Solve both questions : **[14]**

(a) Explain various forces acting on sewer when laid below ground level and running full along with mathematical expressions.

(b) A raw wastewater with the following characteristics is being treated in an activated sludge plant. Given that: BOD_5 of raw wastewater = 220~*gm*/*l*; effluent BOD_5 desired = 20~*mg*/*l*; inflow rate to aeration tank = 100~m^3/*d*; mixed liquor volatile suspended solid concentration = 2000~*mg*/*l*; volume of aeration tank = 500~m^3; water depth in aeration tank = 2.5 m. Determine the:
(i) food to microorganisms (F/M) ratio,
(ii) space loading on aeration tank,
(iii) hydraulic loading on aeration tank and
(iv) detention period in aeration tank.

Q.3 Solve both questions : **[14]**

(a) Explain the various types of solid wastes in detail.

(b) Discuss the types of composition of municipal solid wastes.

Q.4 Solve this question : **[14]**

Compare between separate and combined systems of sewerage. Enumerate the advantages of egg-shaped sewer over circular sewer. Explain the rational formula for estimation of storm run-off along with assumptions. Determine the size of a circular sewer for a discharge of 2.25~m^3/sec and the sewer is to be laid at a gradient of 1 in 7500. Manning's rugosity coefficient = 0.015.

Q.5 Solve both questions : **[14]**

(a) Explain the difference between compaction and size reduction and their importances in solid waste management.

(b) Explain the types, mode of action and applications of equipments used for size reduction and component separation in detail.

Q.6 Solve both questions : **[14]**

(a) List the various materials which can be recovered from the municipal solid waste generated from a typical Indian city.

(b) Describe in brief the various methods employed for the disposal of municipal solid waste.

Q.7 Solve both questions : **[14]**

(a) Explain the importance of the following in the design of sewers: (i) Self cleansing velocity and (ii) Non scouring velocity.

(b) A 2 m deep trickling filter with a diameter of 18 m is operated with a recirculation ratio of 1.5. The raw wastewater flow rate is 2.5 ML/d and the 5-day BOD of the raw wastewater is 210~*mg*/L. Assuming that the primary tank BOD removal efficiency is 30%, compute the hydraulic as well as organic load on the trickling filter.

Q.8 Solve both questions : **[14]**

(a) What do you understand by sludge volume index? Discuss in brief.

(b) Design an oxidation pond using the following data: Population = 5000, Sewage generation = 120~L/*c*/*d*, BOD_5 of sewage = 300~*mg*/*l*, Organic loading = 300~kg/*ha*/*d*.

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Q.9 Write short notes on any four of the following: **[14]**

(a) National river cleaning plans
(b) Roles of Govt. authorities in water supply
(c) Biomedical waste
(d) Recycling of Sewage
(e) 3R
(f) Roles of Govt authorities in control of environmental pollutions

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